

- In each case find the values of expression in Matlab and explain precisely the order in which calculation was performed:-
 - $-2^3 + 9$
 - $\frac{2}{3} * 3$
 - $3 * (\frac{2}{3})$
 - $3 * 4 - 5^2 * 2 - 3$
 - $(\frac{2}{3^2}) * (3 - 4^3)^2$
 - $3 * (3 * 4 - 2 * 5^2 - 3)$
- Write in Matlab code:-
 - $\frac{(2+6i)}{(9i)(1+i)}$
 - Find the absolute value and angle in Matlab of part (a).
- Given array $A = [2, 4, 1; 6, 7, 2; 3, 5, 9]$ provide the commands needed to:-
 - Assign the first row of A to vector called x .
 - Assign the last two row of A an array called y .
 - Compute the sum over the column of A .
- Let $X = [3, 2, 6, 8]^T$ and $Y = [4, 1, 3, 5]^T$ add X and Y then find sum of elements of resulting vector.
- Let $A = [1, 2, 3, 4; 5, 6, 7, 8; 1, 1, 2, 3]$ then write in Matlab:-
 - To extract a sub-matrix B consisting of row 1 and 3, column 2 and 4.
 - Inter change row 1 and 3.
 - Find $A(:)$. Explain what it does?
 - Delete 2^{nd} column of A and delete 3^{rd} row of A .
 - Remove the second column.
- Suppose vectors a and b are defined as follows:

$$a = [2, -1, 5, 0];$$

$$b = [3, 2, -1, 4];$$
 Evaluate by hand the vector c in the following statements. Check your answers in Matlab.
 - $c = a - b$.
 - $c = +a - 3b$.
 - $c = 2 * a + a.^b$.
 - $c = b./a$.
 - $c = b.a$.
 - $c = 2 * b/3. * a$.
 - $c = b * 2. * a$.
- Given a script for animating the Mexican hat problem.


```
[x,y] = meshgrid (-8 : 0.5 : 8);
r = sqrt( x.^2 + y.^2 ) + eps;
z = sin(r)./r;
mesh(z);
```

 type this into the editor, save it,and execute it.
- Try to avoid using unnecessary brackets in an expressions. Can you spot the errors in the following expression?

$$(2(3 + 4)/(5 * (6 + 1))^2$$
- Use Matlab array operations to do the following :
 - Add 1 to each element of the vector $[2, 3, -1]$.
 - Multiply each element of the vector $[1, 4, 8]$ by 3.
 - Find the array product of the two vector $[1, 2, 3]$ and $[0, -1, 1]$.
 - square each element of the vector $[2, 3, 1]$.